YIWEN ZHANG

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EDUCATION

Stanford University School of Medicine *Stanford, CA*
CAMPEP Medical Physics Certificate Program *Sept. 2026 – Present*

California Institute of Technology *Pasadena, CA*
Ph.D. in Physics *2019 – 2025*

- Advisor: Prof. Kathryn M. Zurek (GPA: 4.3/4.0)
- Thesis: *Quantum Gravity and Laser Interferometry: Towards Observable Predictions*

University of Wisconsin–Madison *Madison, WI*
B.S. in Applied Mathematics, Physics & Engineering *2015 – 2019*

- GPA: 3.97/4.0

RESEARCH EXPERIENCE

AI Navigation Agents Development, Caltech *Pasadena, CA*
Reinforcement Learning & Robotics Navigation *Aug. 2025 – June 2026*

- Designed AI navigation agents for quadruped robots using imitation and reinforcement learning; benchmarked via large-scale physics simulations in Python

Ph.D. Research, Caltech *Pasadena, CA*
High Energy Physics / Quantum Gravity Group *Sept. 2019 – June 2025*

- Developed stochastic models and applied signal detection theory and Fourier analysis to extract weak quantum gravity signals from noisy backgrounds
- Built large-scale numerical simulations demonstrating signal detectability above background noise by orders of magnitude; led multiple projects to peer-reviewed publication

PROJECTS

Cloud-Based Clinical Data-Collection & Case-Management Platform *2025 – Present*

- Co-developed a cloud-based mobile app for dental charting and treatment planning, deployed in special-needs clinics in Mexico and Taiwan (~100 patients per site)
- Digitized and standardized paper records; added multilingual support (Spanish, English, Traditional Chinese)

Smart Wildlife Camera Trap *2025 – Present*

- Designed AI-powered wildlife detection system using computer vision and low-power edge inference hardware

INDUSTRY EXPERIENCE

Quantitative Analytics Model Development, Goldman Sachs *New York, NY*
June – Aug. 2025

- Built and improved quantitative models for portfolio risk analysis and structured product optimization using linear and quadratic programming; achieved 40% runtime reduction

SKILLS

Physics & Math: Quantum Mechanics, Stochastic Processes, Fourier Analysis, Signal Detection Theory, Statistical Inference, Differential Equations

Programming: Python, NumPy, SciPy, Pandas, Matplotlib, Scikit-learn, PyTorch, JAX, Flutter/Dart, GCP, Mathematica, LaTeX

Machine Learning & Artificial Intelligence: Deep RL, Imitation Learning, Computer Vision, Object Detection

PEER-REVIEWED PUBLICATIONS

1. **Y. Zhang**, K. M. Zurek, "Stochastic description of near-horizon fluctuations in Rindler-AdS," *Physical Review D*, 108, 066002 (2023) **[First Author]**
2. M. W. Bub, T. He, P. Mitra, **Y. Zhang**, K. M. Zurek, "Quantum mechanics of a spherically symmetric causal diamond in Minkowski spacetime," *Physical Review Letters*, 134, 121501 (2025) **[Co-First with M. W. Bub]**
3. S-E. Bak, C. Keeler, **Y. Zhang**, K. M. Zurek, "Rindler fluids from gravitational shockwaves," *Journal of High Energy Physics*, 05 (2024) 331 **[Co-First with S-E. Bak]**
4. M. W. Bub, Y. Chen, Y. Du, D. Li, **Y. Zhang**, K. M. Zurek, "Quantum gravity background in next-generation gravitational wave detectors," *Physical Review D*, 108, 064038 (2023)

**Authors listed alphabetically per field convention.*

WORKSHOP PAPERS

1. **Y. Zhang**, E. Zeng, J. Lee, T. Y. Yu, "An AI Scientist Guided by Encoded Human Research Preferences: A Case Study in Quadruped Neural Navigation," *ICML Workshop*, (2026) **[Co-First with J. Lee and E. Zeng]**